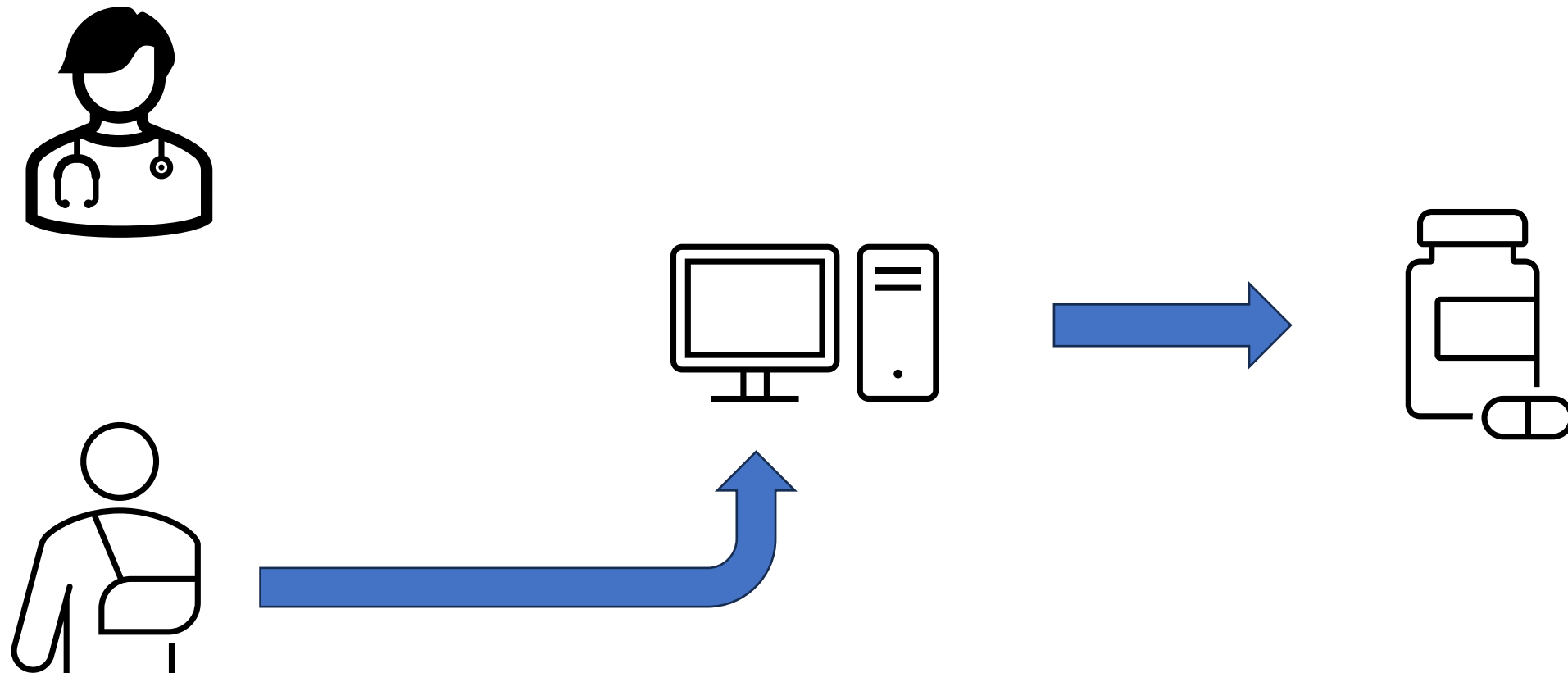


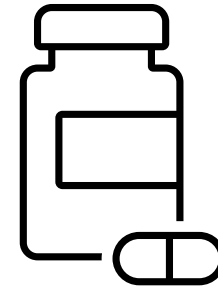
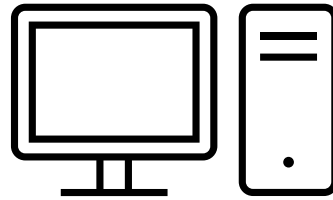
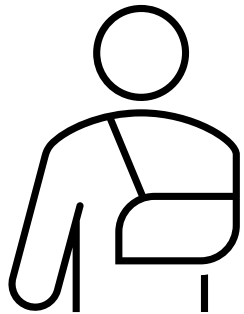
Learning Performance Maximizing Ensembles with Explainability Guarantees

Vincent Pisztor

Motivation



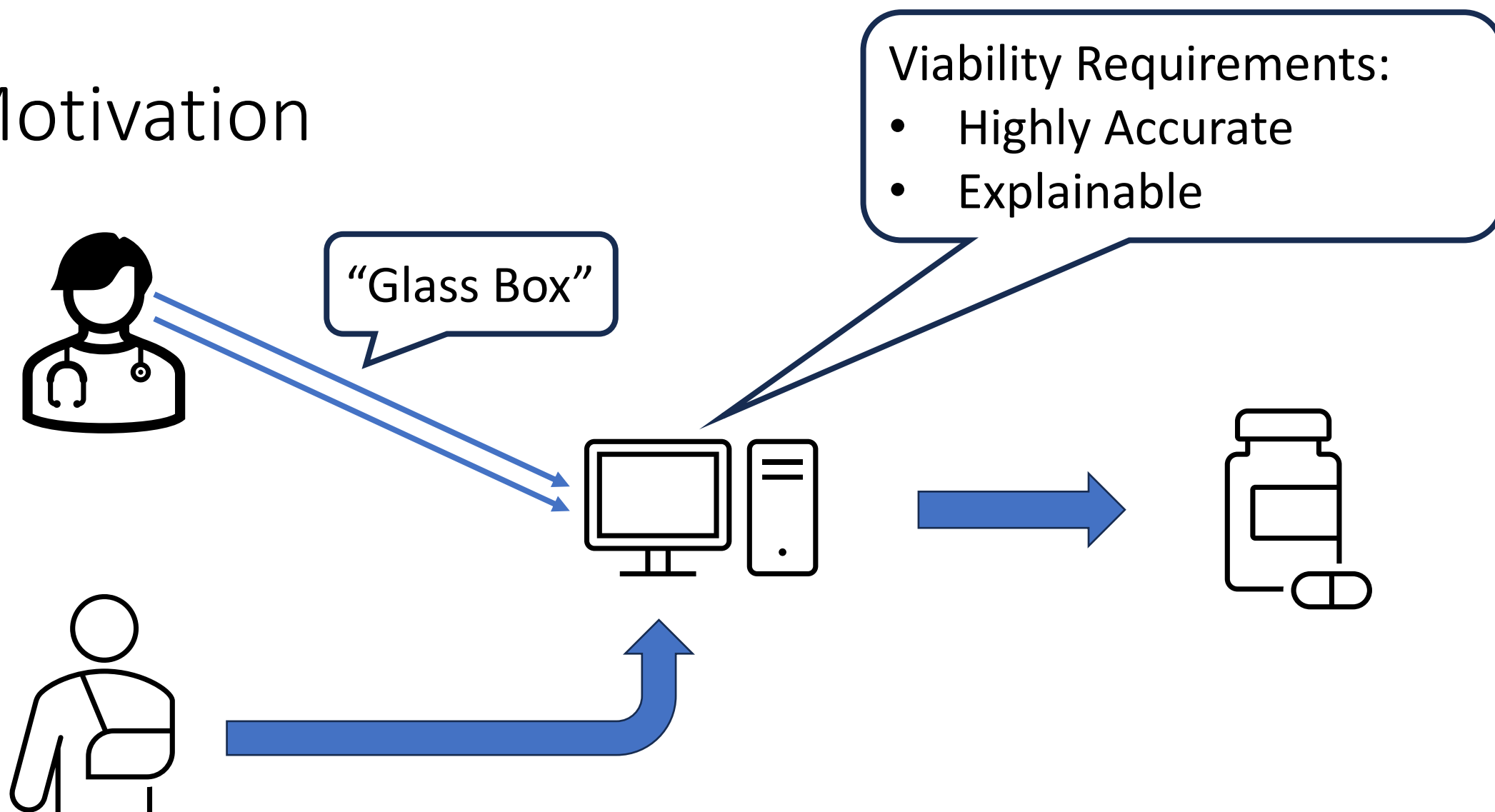
Motivation



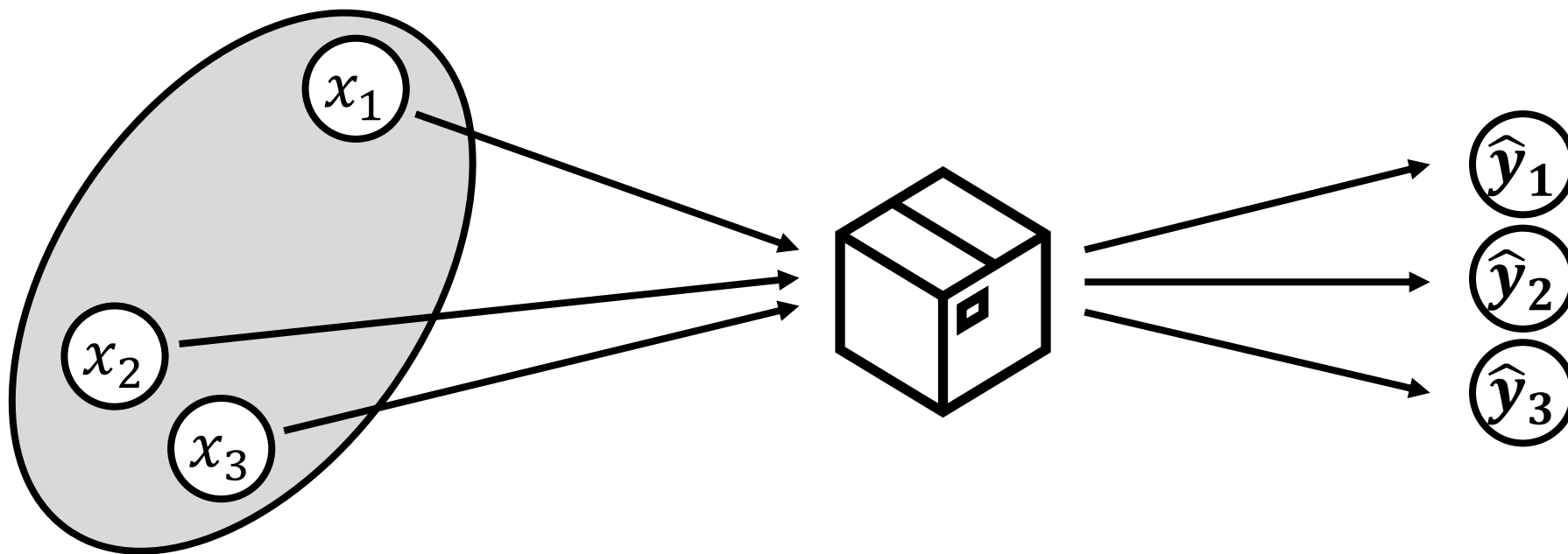
Viability Requirements:

- Highly Accurate
- Explainable

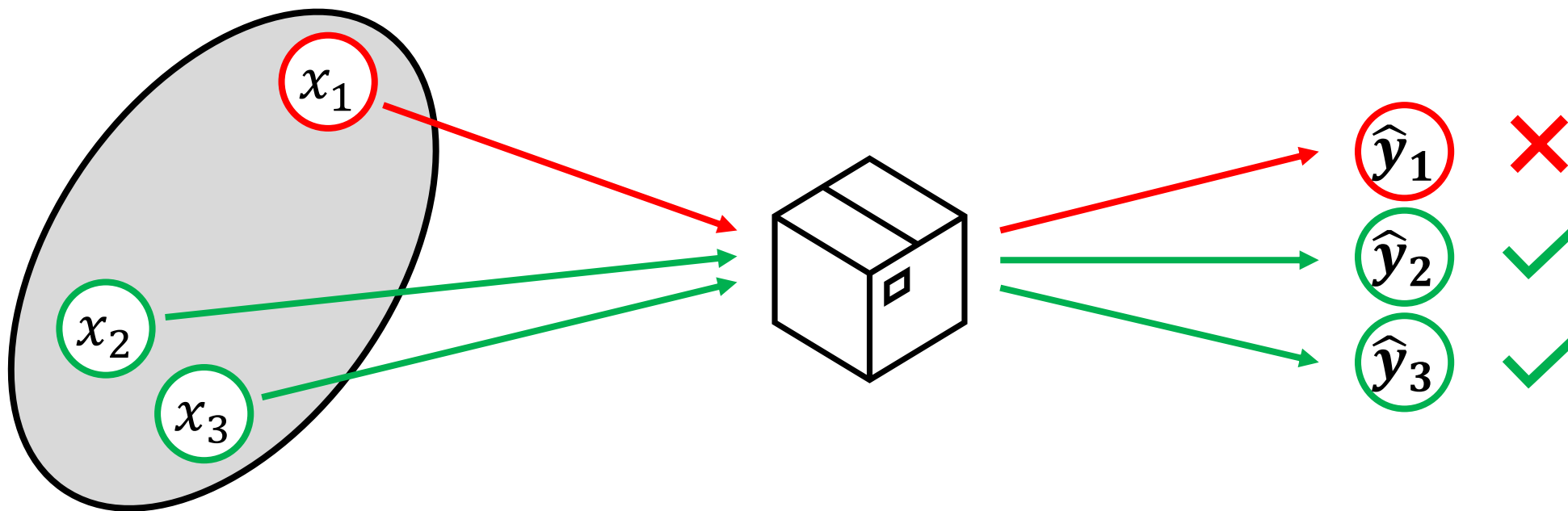
Motivation



Setting

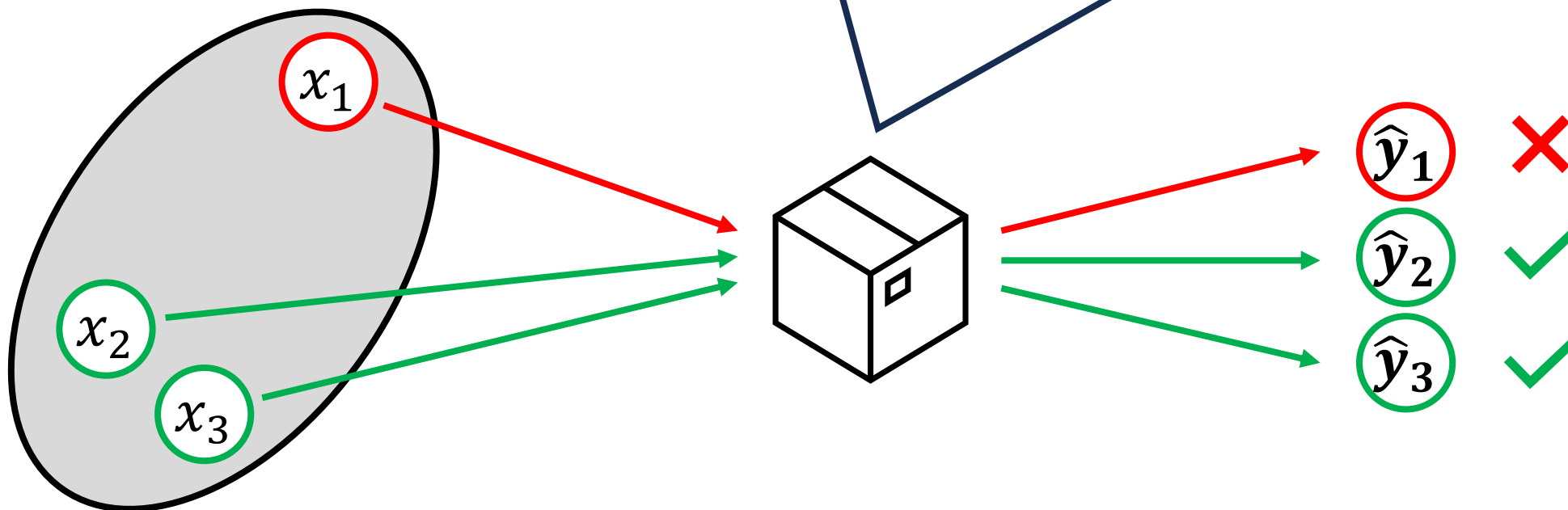


Setting



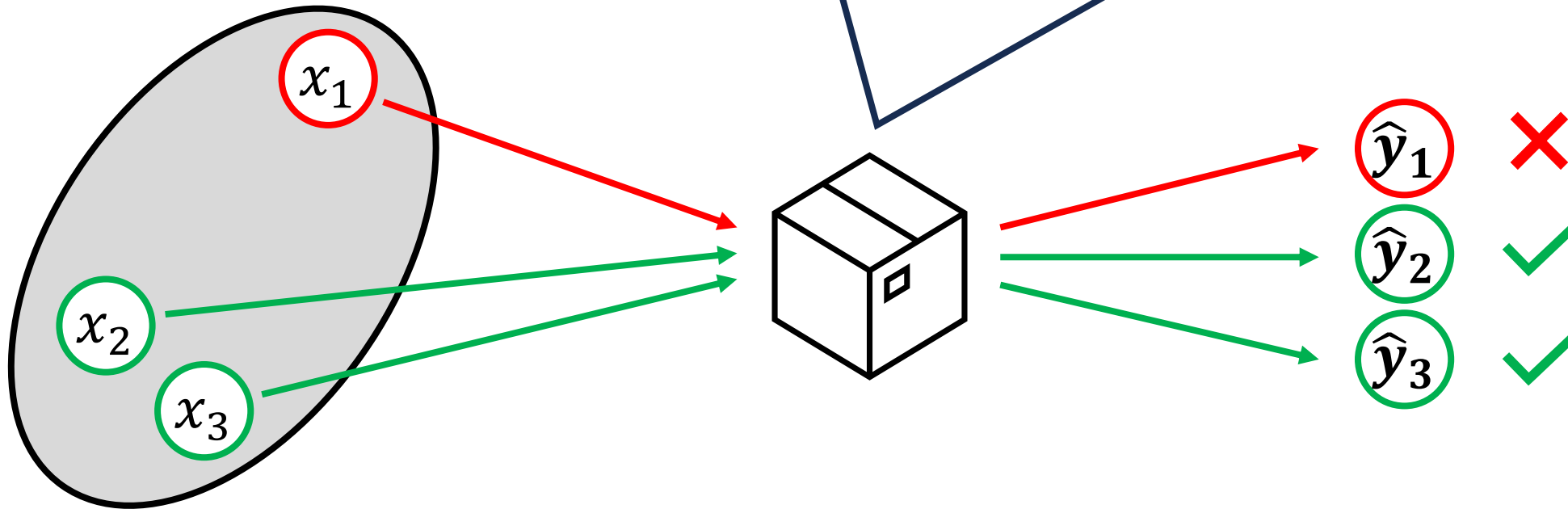
Setting

- Glass Box Predicts *All* Observations
- Doesn't Meet Performance Threshold



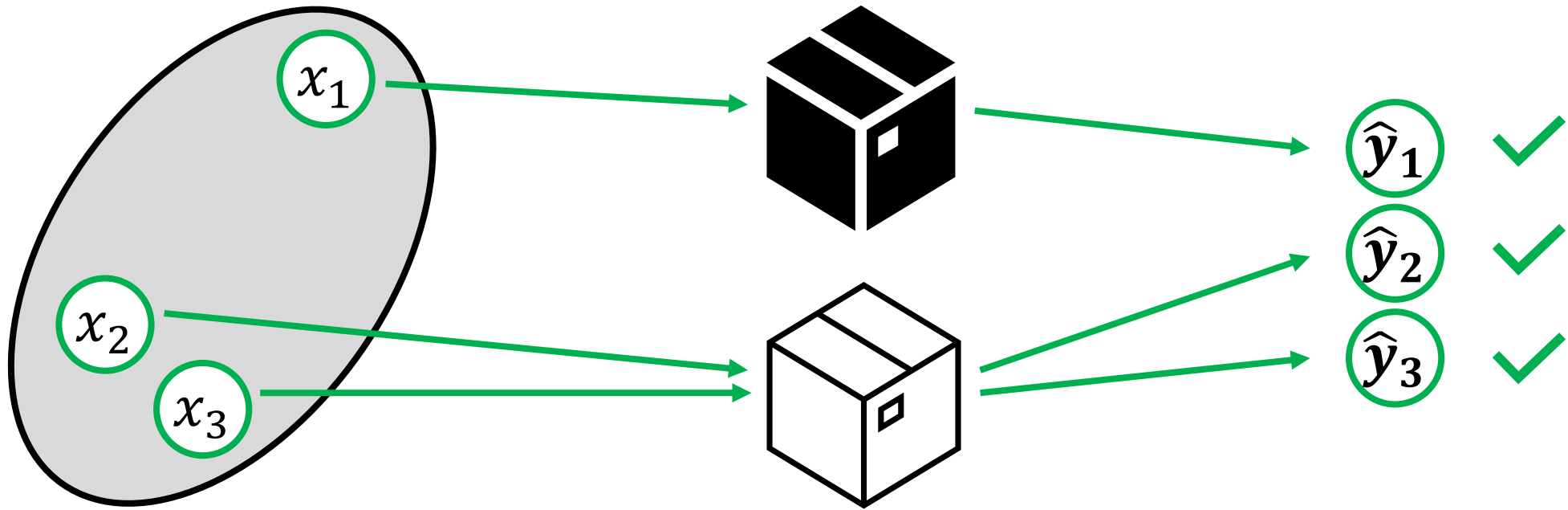
Setting

- Glass Box Predicts *All* Observations
- Doesn't Meet Performance Threshold

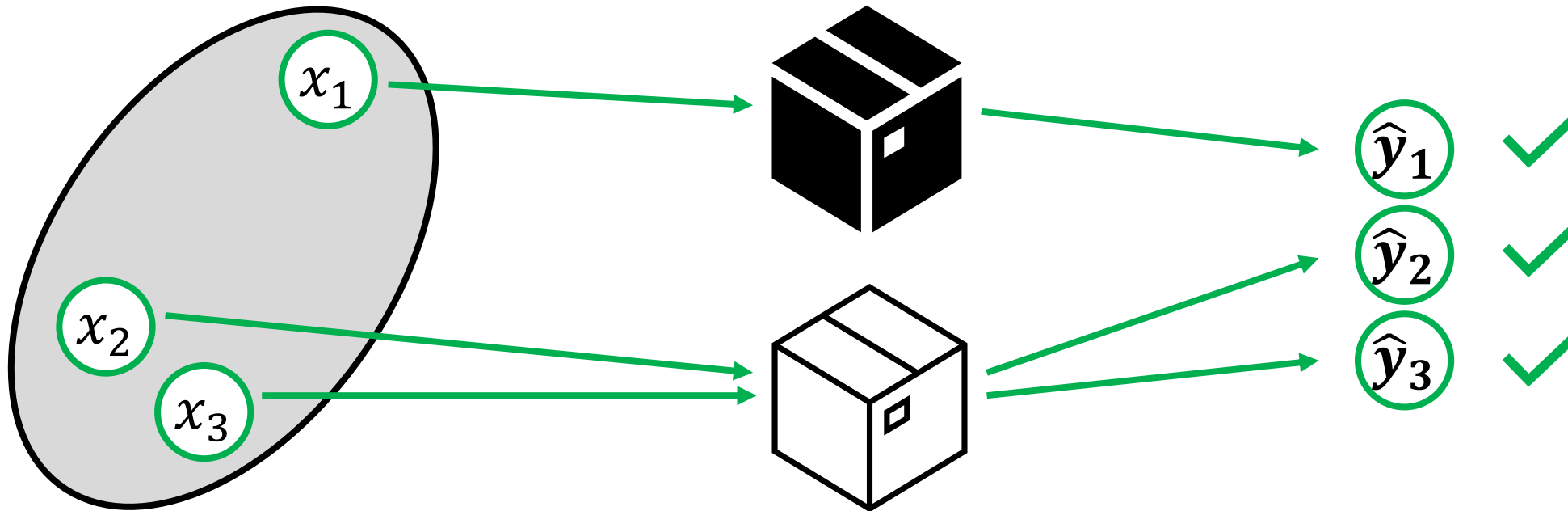


If a glass box alone can't meet both requirements, we can relax the explainability requirement

Setting – Partial Explainability



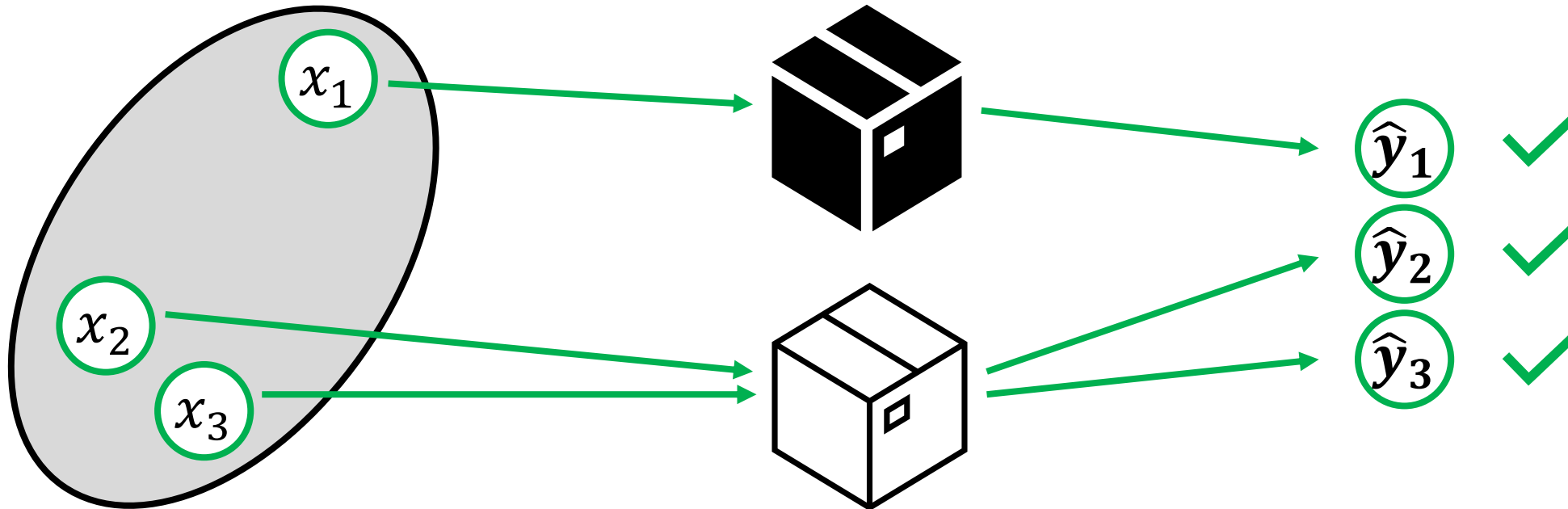
Setting – Partial Explainability



$$\text{Transparency/Explainability Level} = q = \frac{\# \text{ Glass Box Predictions}}{\# \text{ Total Predictions}}$$

Setting – Partial Ex

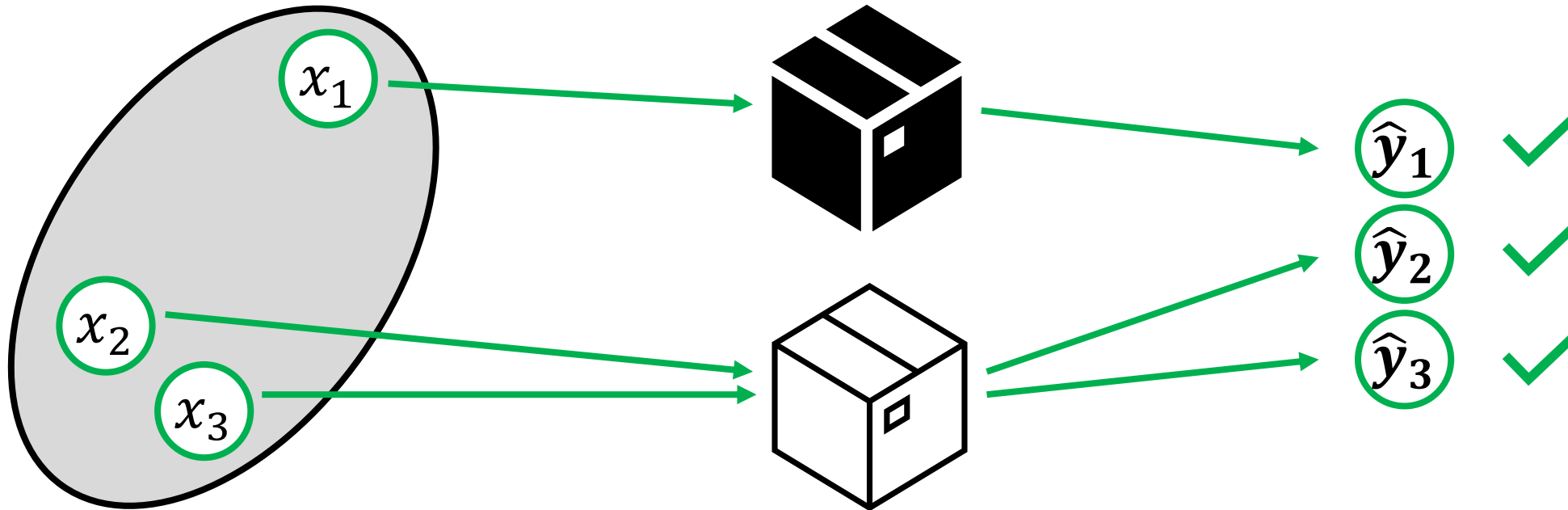
- Glass Box Predicts *Most* Observations
- Black Box Predicts *Some* Observations
- Ensemble Meets Performance Threshold



$$\text{Transparency/Explainability Level} = q = \frac{\# \text{ Glass Box Predictions}}{\# \text{ Total Predictions}}$$

Setting – Partial Ex

- Glass Box Predicts *Most* Observations
- Black Box Predicts *Some* Observations
- Ensemble Meets Performance Threshold



But how to decide which observation gets allocated to which model?

Optimal Allocation Step-by-Step

For a given q :

- 1) Define conditions which an optimal allocator should meet
- 2) Define an observation-level metric that quantifies how much an observation's allocation to, say the glass box, would improve the optimality conditions
- 3) Using this metric, evaluate each observation's glass box allocation desirability level
- 4) Rank all observations according to their glass-box-allocation-desirability
- 5) Allocate the q percent of observations with the highest glass-box-allocation-desirability to the glass box for prediction

Optimal Allocation Conditions

An optimal allocation is such that, at a given q :

- 1) There exist no allocations for which the ensemble is more accurate
- 2) There exist no allocations for which the glass box is more accurate on the subset of observations it has been allocated
- 3) Among those allocations which meet conditions (1) and (2), there exist none for which the loss of the ensemble is lower

Optimal Allocation Conditions (specific)

An optimal allocation is such that, at a given q :

- 1) There exist no allocations for which the ensemble **is more accurate**
- 2) There exist no allocations for which the glass box **is more accurate** on the subset of observations it has been allocated
- 3) Among those allocations which meet conditions (1) and (2), there exist none for which the **loss** of the ensemble is **lower**

Optimal Allocation Conditions (general)

An optimal allocation is such that, at a given q :

- 1) There exist no allocations for which the ensemble **has higher sufficient performance**
- 2) There exist no allocations for which the glass box **has higher sufficient performance** on the subset of observations it has been allocated
- 3) Among those allocations which meet conditions (1) and (2), there exist none for which the **absolute performance** of the ensemble is **higher**

Optimal Allocation Step-by-Step

For a given q :

- ✓ Define conditions which an optimal allocator should meet
- 2) Define an observation-level metric that quantifies how much an observation's allocation to, say the glass box, would improve the optimality conditions
- 3) Using this metric, evaluate each observation's glass box allocation desirability level
- 4) Rank all observations according to their glass-box-allocation-desirability
- 5) Allocate the q percent of observations with the highest glass-box-allocation-desirability to the glass box for prediction

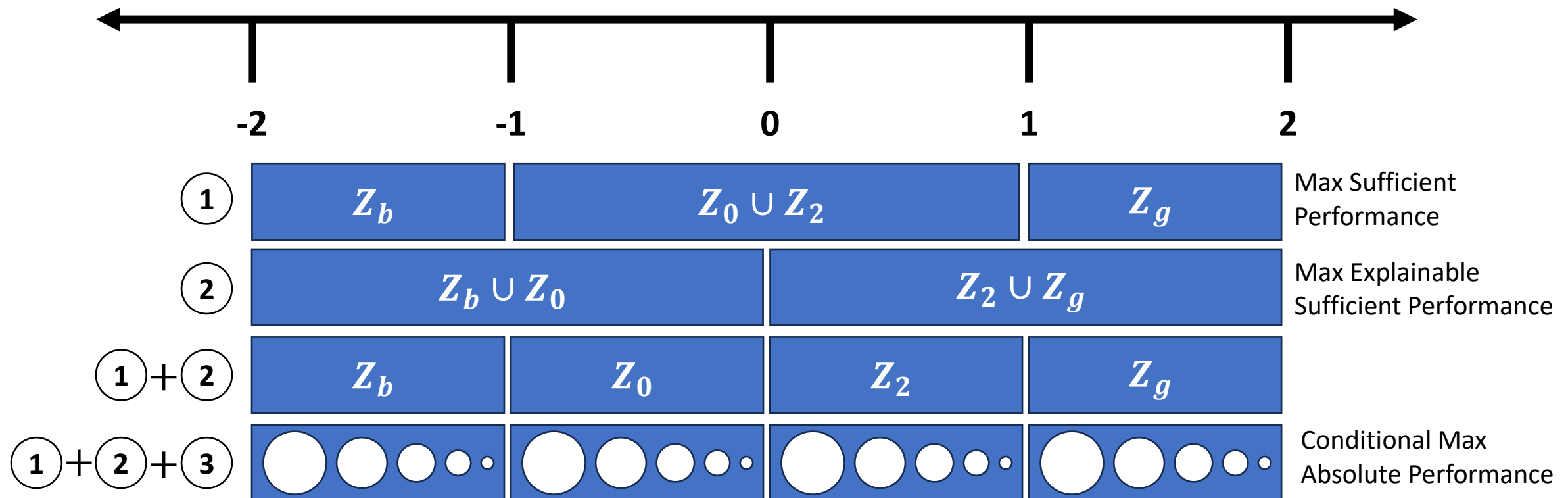
Glass Box Allocation Desirability

$$\tilde{r}(x, y) = 2s_g(x, y) - s_b(x, y) - \sigma(\ell(g(x), y) - \ell(b(x), y))$$

- Glass Box: $g: \mathcal{X} \rightarrow \mathcal{Y}$
- Black Box: $b: \mathcal{X} \rightarrow \mathcal{Y}$
- Sigmoid: $\sigma: \mathcal{X} \rightarrow (0,1)$
- Loss: $\ell: \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}$
- Sufficiency: $s_f: \mathcal{X} \times \mathcal{Y} \rightarrow \{0,1\}$
- Sufficiency Sets:
 - $Z_0 = \{(x, y): s_g = 0, s_b = 0\} \xrightarrow{\tilde{r}} (-1, 0)$
 - $Z_g = \{(x, y): s_g = 1, s_b = 0\} \xrightarrow{\tilde{r}} (1, 2)$
 - $Z_b = \{(x, y): s_g = 0, s_b = 1\} \xrightarrow{\tilde{r}} (-2, -1)$
 - $Z_2 = \{(x, y): s_g = 1, s_b = 1\} \xrightarrow{\tilde{r}} (1, 0)$

Glass Box Allocation Desirability

$$\tilde{r}(x, y) = 2s_g(x, y) - s_b(x, y) - \sigma(\ell(g(x), y) - \ell(b(x), y))$$



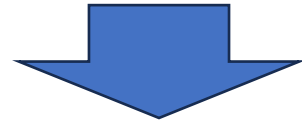
Optimal Allocation Step-by-Step

For a given q :

- ✓ Define conditions which an optimal allocator should meet
 - ✓ Define an observation-level metric that quantifies how much an observation's allocation to, say the glass box, would improve the optimality conditions
 - ✓ Using this metric, evaluate each observation's glass box allocation desirability level
 - ✓ Rank all observations according to their glass-box-allocation-desirability
- 5) Allocate the q percent of observations with the highest glass-box-allocation-desirability to the glass box for prediction

Optimal Allocator

$$\tilde{r}(x, y) = 2s_g(x, y) - s_b(x, y) - \sigma(\ell(g(x), y) - \ell(b(x), y))$$



Normalize

$$r(x, y) = \frac{\text{rank}_{\mathcal{D}^n} \tilde{r}(x, y)}{n}$$



Select the q most desirable

$$a_q(x, y) = \mathbb{I}\{r(x, y) > 1 - q\}$$

Optimal Allocation Step-by-Step

For a given q :

- ✓ Define conditions which an optimal allocator should meet
- ✓ Define an observation-level metric that quantifies how much an observation's allocation to, say the glass box, would improve the optimality conditions
- ✓ Using this metric, evaluate each observation's glass box allocation desirability level
- ✓ Rank all observations according to their glass-box-allocation-desirability
- ✓ Allocate the q percent of observations with the highest glass-box-allocation-desirability to the glass box for prediction

Optimal Allocation

What about for another transparency level?
Do we need to redo everything?

For a given q :

- ✓ Define conditions which an optimal allocator should meet
- ✓ Define an observation-level metric that quantifies how much an observation's allocation to, say the glass box, would improve the optimality conditions
- ✓ Using this metric, evaluate each observation's glass box allocation desirability level
- ✓ Rank all observations according to their glass-box-allocation-desirability
- ✓ Allocate the q percent of observations with the highest glass-box-allocation-desirability to the glass box for prediction

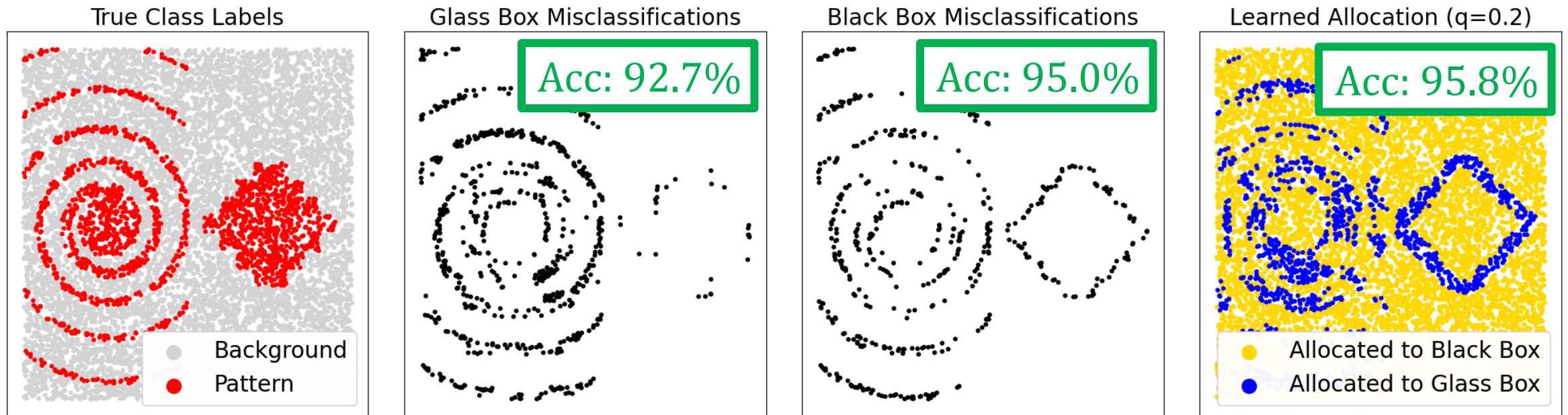
Learning Optimal Allocations for any q

- Desirability ranking does not depend on the transparency level q and is in fact optimal for any q
- Learn the ranking, not the allocation
- For each q , simply change the allocation cutoff value
- Train once, obtain optimal allocations for any q

$$\tilde{r}(x, y) = 2s_g(x, y) - s_b(x, y) - \sigma(\ell(g(x), y) - \ell(b(x), y))$$

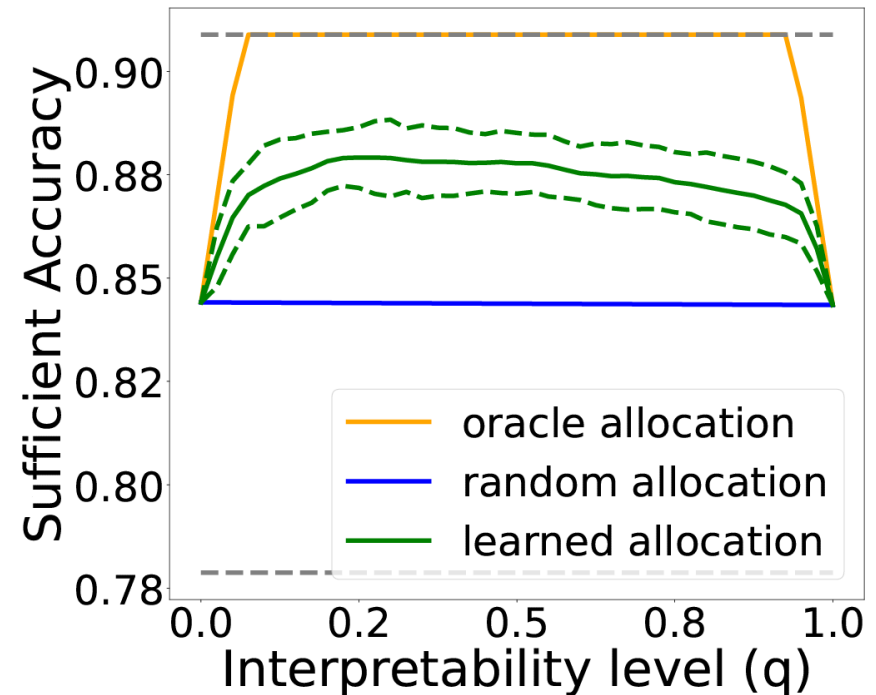
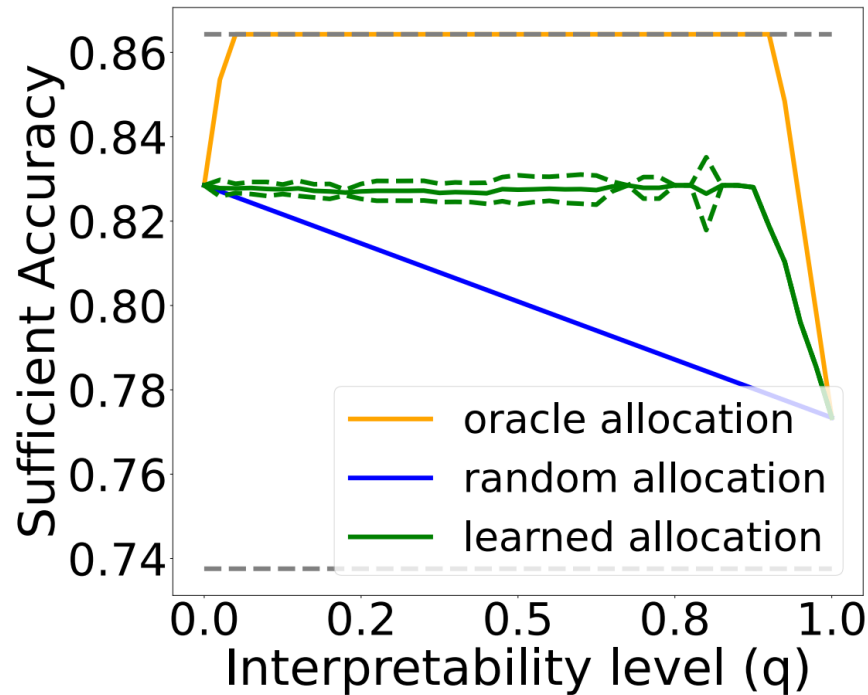
$$a_q(x, y) = \mathbb{I}\{r(x, y) > 1 - q\}$$

The Ideal Scenario – Complementary Models



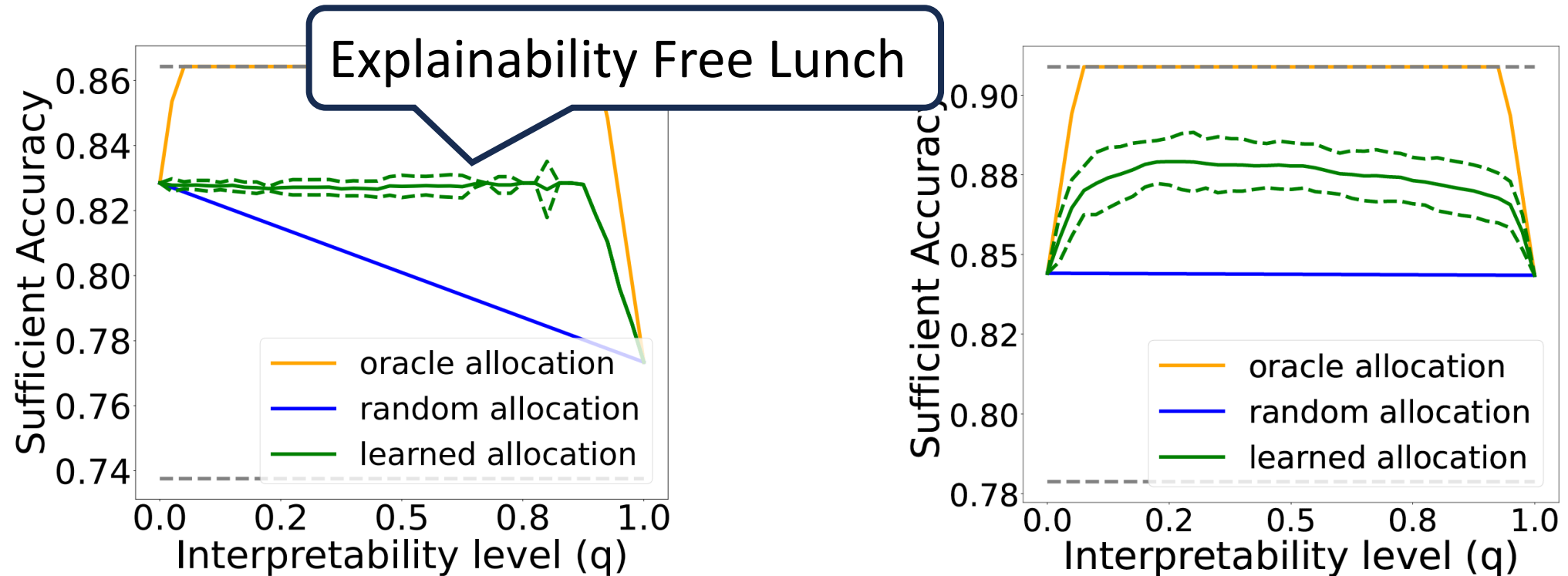
The areas of expertise (the diamond pattern for the glass box and the spiral pattern for the black box model) are complementary. Allowing the resulting allocated ensemble to improve performance over both component models while also providing explainability.

How About on Real Data?



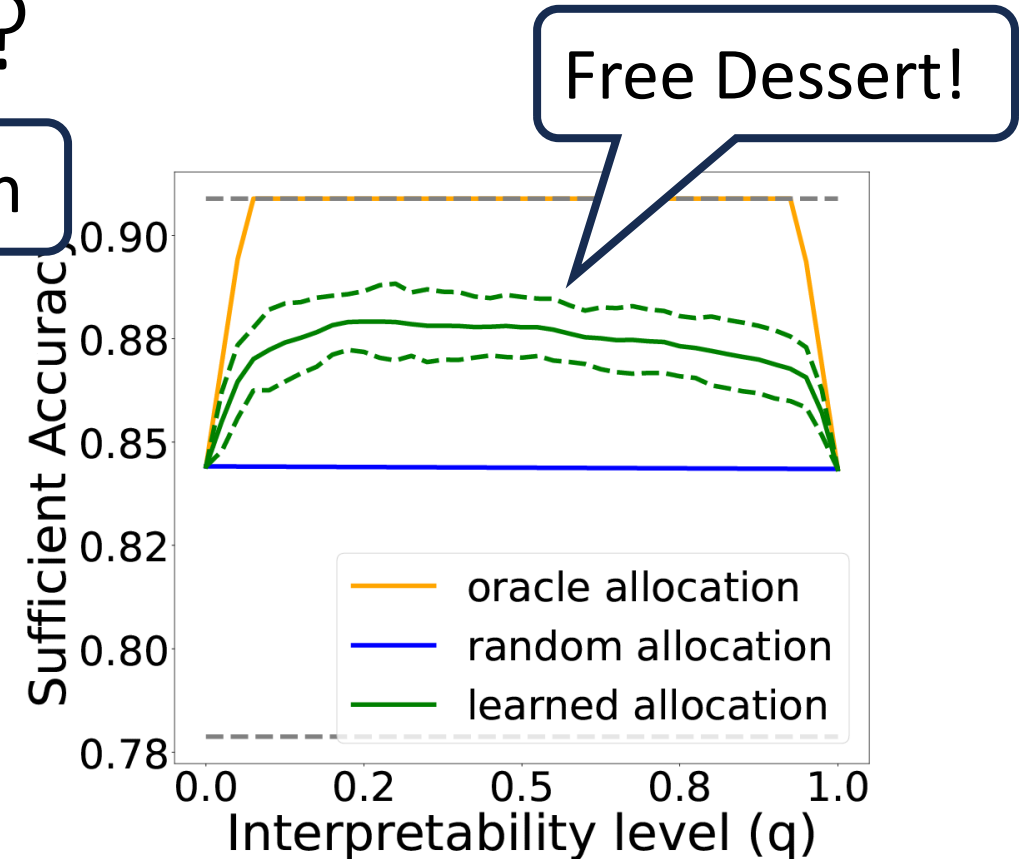
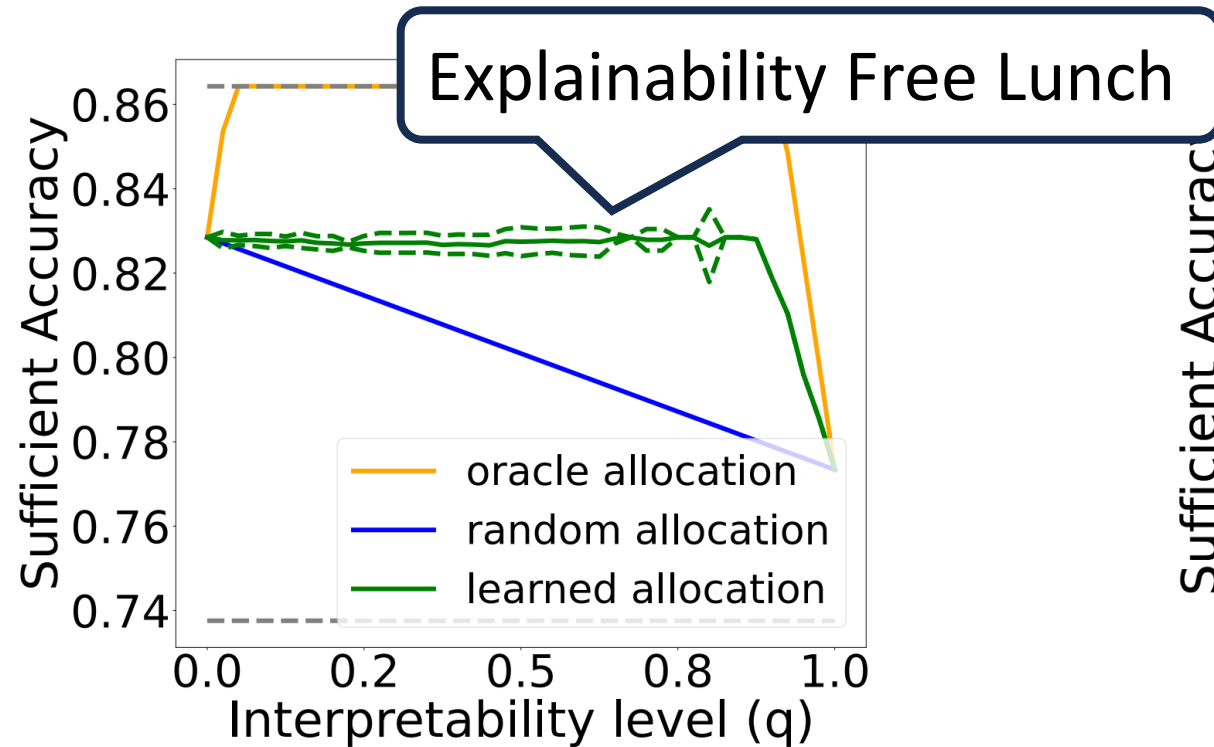
Learned allocations are found to consistently maintain ensemble performance at very high explainability levels (explaining 74% of observations on average) and in some cases even outperform both the component models while improving explainability

How About on Real Data?



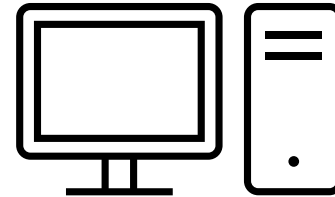
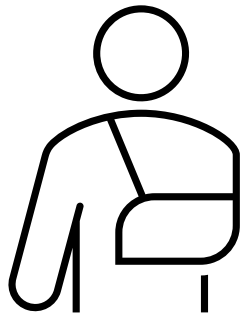
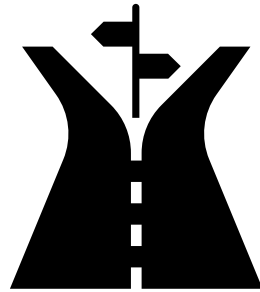
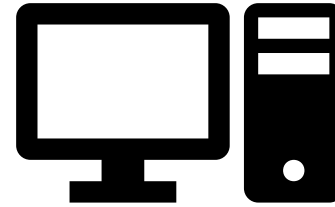
Learned allocations are found to consistently maintain ensemble performance at very high explainability levels (explaining 74% of observations on average) and in some cases even outperform both the component models while improving explainability

How About on Real Data?

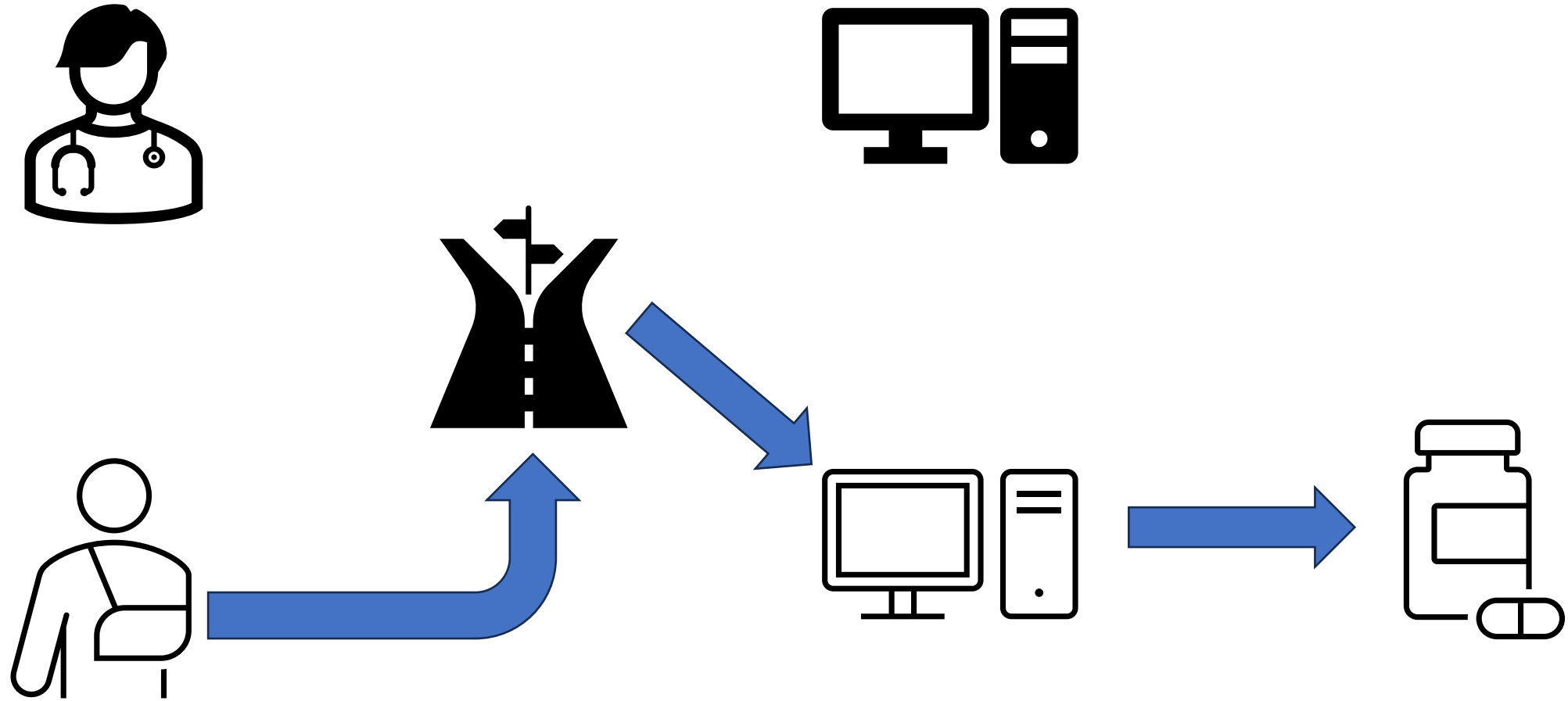


Learned allocations are found to consistently maintain ensemble performance at very high explainability levels (explaining 74% of observations on average) and in some cases even outperform both the component models while improving explainability

Conclusion

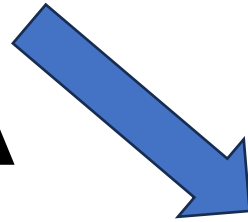
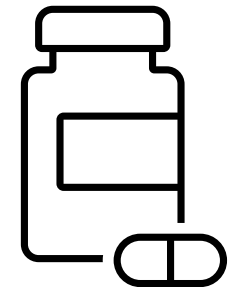
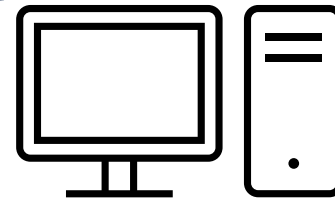
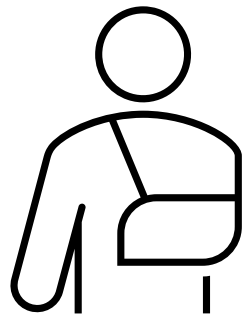
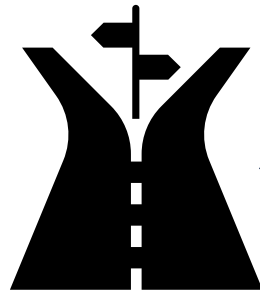
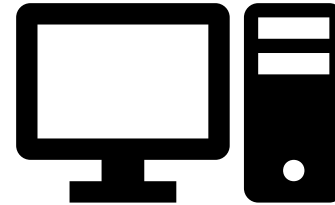


Conclusion



Conclusion

Great! An accurate glass box diagnosis,
let me check the reasoning...

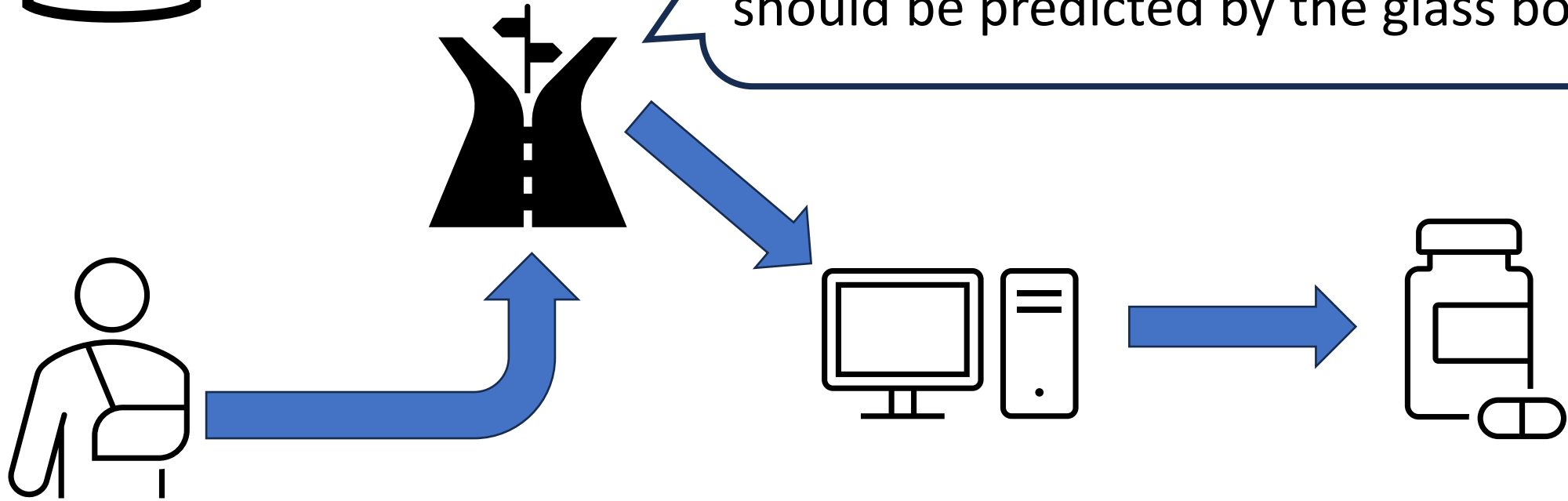


Conclusion

Great! An accurate glass box diagnosis, let me check the reasoning...



Oh! A r of 94%, a very high desirability score, the allocator really thinks this case should be predicted by the glass box

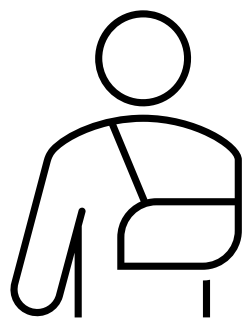
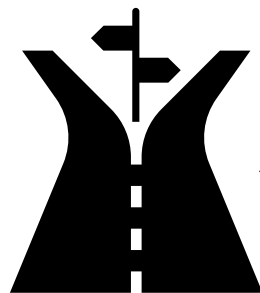


Conclusion

Great! An accurate glass box diagnosis, let me check the reasoning...



Oh! A r of 94%, a very high desirability score, the allocator really thinks this case should be predicted by the glass box



Hm.. I see, the estimated sufficiency class is Z_g , so the black box prediction is likely incorrect

